Name of the Student: Rohan Dwivedi Scholar Number: 24212011118

मौलाना आजाद राष्ट्रीय प्रौद्योगिकी संस्थान, भोपाल- 462003

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL- 462003

संगणक विज्ञान एवं अभियांत्रिकी विभाग

Department of Computer Science and Engineering

Mid-Term Examination, Nov 2024

Course: M.Tech (CN)
Subject: TCP-IP
Date: 19-11-2023
Time: 09:30AM to 10:30AM

Semester: I

Branch: CSE
Subject Code: FCN 511
Max Marks: 20

Notes: Take necessary justification when required and clearly explain all the steps.

Q. No.	Question	Marks	COs
Q1.	Explain why CSMA/CD is unsuitable for wireless networks, specifically addressing the issues related to signal propagation and collision detection in a shared medium. How does CSMA/CA address these challenges, and what mechanisms are used to minimize collisions in wireless communications?	4	CO1
Q2.	A service provider has multiple Class C networks that need to be aggregated to simplify routing. Explain how Classless Inter-Domain Routing (CIDR) is used for aggregation in this scenario. Given four networks (192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24, 192.168.4.0/24), provide the supernet mask and explain the advantages of using this supernet.	4	CO1 & CO2
Q3.	The IPv4 header includes several fields that support fragmentation. Explain the roles of the identification, flags, and fragmentation offset fields in the fragmentation process. Given an IP packet of size 4000 bytes (including 20 bytes of header) being sent over a network with an MTU of 1500 bytes, demonstrate how the packet will be fragmented. Provide details of each fragment, including their sizes and header fields.	4	CO1 & CO2
Q4.	In a network using the Distance Vector Routing Protocol, discuss the "Count to Infinity" problem. Use an example with three routers (A, B, C) to demonstrate how this issue arises. Explain how split horizon and poison reverse mitigate this problem.	4	CO1 & CO2
Q5.	Prove why the Selective Repeat (SR) protocol's send window size must be less than or equal to 2^{m-1}, where m is the number of bits used for sequence numbers. Use a suitable example to illustrate your explanation.	4	CO1 & CO2

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Name of the Student..... Scholar Number.....

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संगणक विज्ञान एवं अभियांत्रिकी विभाग

Department of Computer Science and Engineering

End-Term Examination, Dec 2024

Course: M.Tech (CN) **Semester:** I **Branch:** CSE

Subject: TCP-IP **Subject Code:** FCN 511

Date: 19-12-2024 **Max Marks: 40**

Time: 09:30AM to 11:30AM

Notes: Take necessary justification when required and clearly explain all the steps.

Q. No.	Question	Marks	COs
Q1.	Assume the ethernet LAN is configured with CSMA/CD using bus topology. If the stations are using frames of size 500 bits and the datarate offered by a cable is 50 mbps (megabits per second). The signal speed inside the cable is 2,50,000 km/sec. The stations are connected uniformly with a spacing of 10 mts between the adjacent ones. Calculate the maximum number stations that can be connected?	5	CO1
Q2.	In a network using the Distance Vector Routing Protocol, discuss the "Count to Infinity" problem. Use an example with three routers (A, B, C) to demonstrate how this issue arises. Explain how split horizon and poison reverse mitigate this problem.	5	CO1 & CO2
Q3.	For reliable frame transmission, a sender uses Stop-and-Wait ARQ protocol. Frames are 500 bytes in size, and the sender's transmission rate is 40 Kbps (1Kbps = 1000 bits per second). The size of an acknowledgement is 50 bytes, and transmission rate of the receiver is 2 Kbps. The propagation delay of the channel is 100 milliseconds. Calculate the throughput assuming no frames are lost.	5	CO1 & CO2
Q4.	Consider an IPv4 packet with option field(s) of 20 bytes. Calculate the checksum field for the IPv4 header when *service type* field is 0, *total length* field is 60, *identification* field is 1, *do not fragment* (*D*) bits is set to 1, *TTL* is 4, the higher layer protocol is TCP, initial value of the checksum is set to 0 at the sender, the *source address* is 5.11.7.5 and the *destination address* is 10.10.8.4.	5	CO1 & CO2
Q5.	Prove why the Selective Repeat (SR) protocol's send window size must be less than or equal to 2^{m-1}, where m is the number of bits used for sequence numbers. Use a suitable example to illustrate your explanation.	5	CO1 & CO2
Q6.	What challenges in IP communication require mechanisms for error reporting and diagnostics? How does ICMP address these challenges? Focus specifically on the functionality of the Destination Unreachable and Time Exceeded ICMP messages. Provide examples of scenarios where these messages are generated and analyze their importance in ensuring reliable packet delivery and effective troubleshooting in IP networks.	5	CO1 & CO2
Q7.	Describe the structure of an SCTP packet, including the roles of common control and data chunks. Provide a detailed explanation of SCTP's four-way handshake process for establishing a connection and its mechanism for handling association termination.	5	CO3
Q8.	Explain the primary purpose of the Dynamic Host Configuration Protocol (DHCP). Describe the four-step process of IP address allocation (Discover, Offer, Request, Acknowledge) in detail, including the role of DHCP servers and clients. Provide an example scenario where DHCP simplifies network configuration.	5	CO3

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